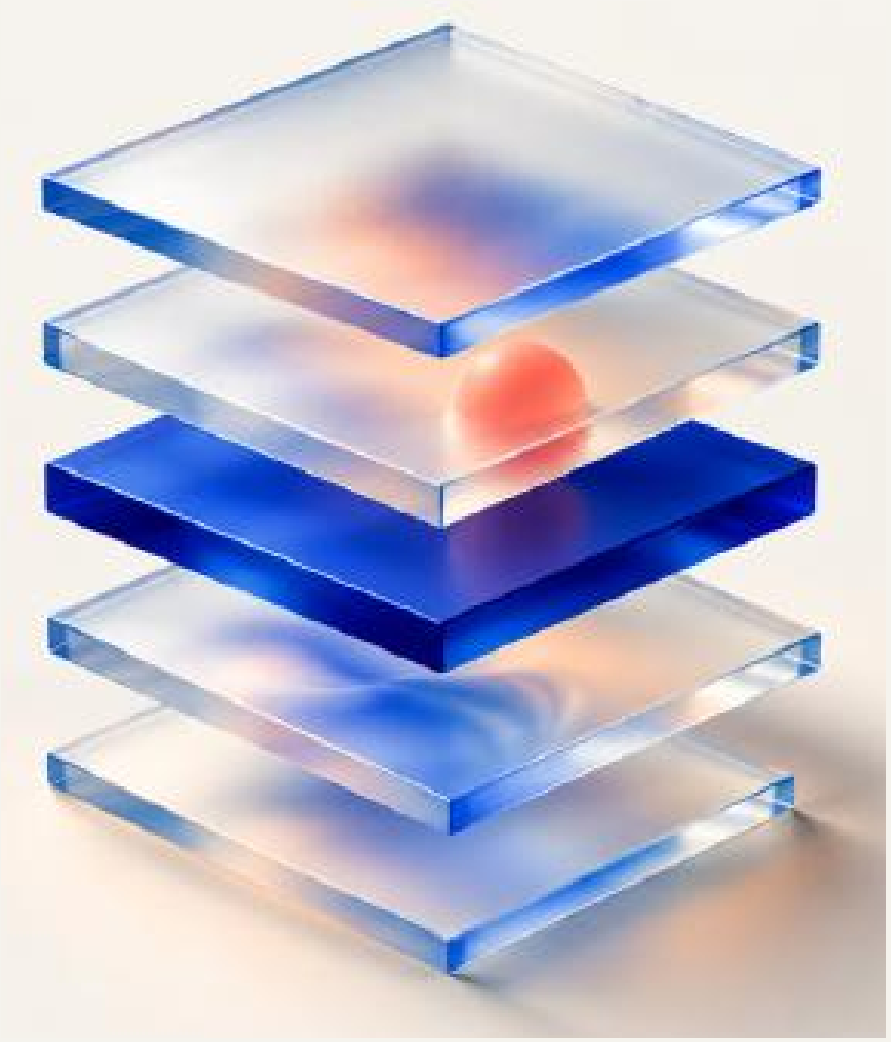
Blackout

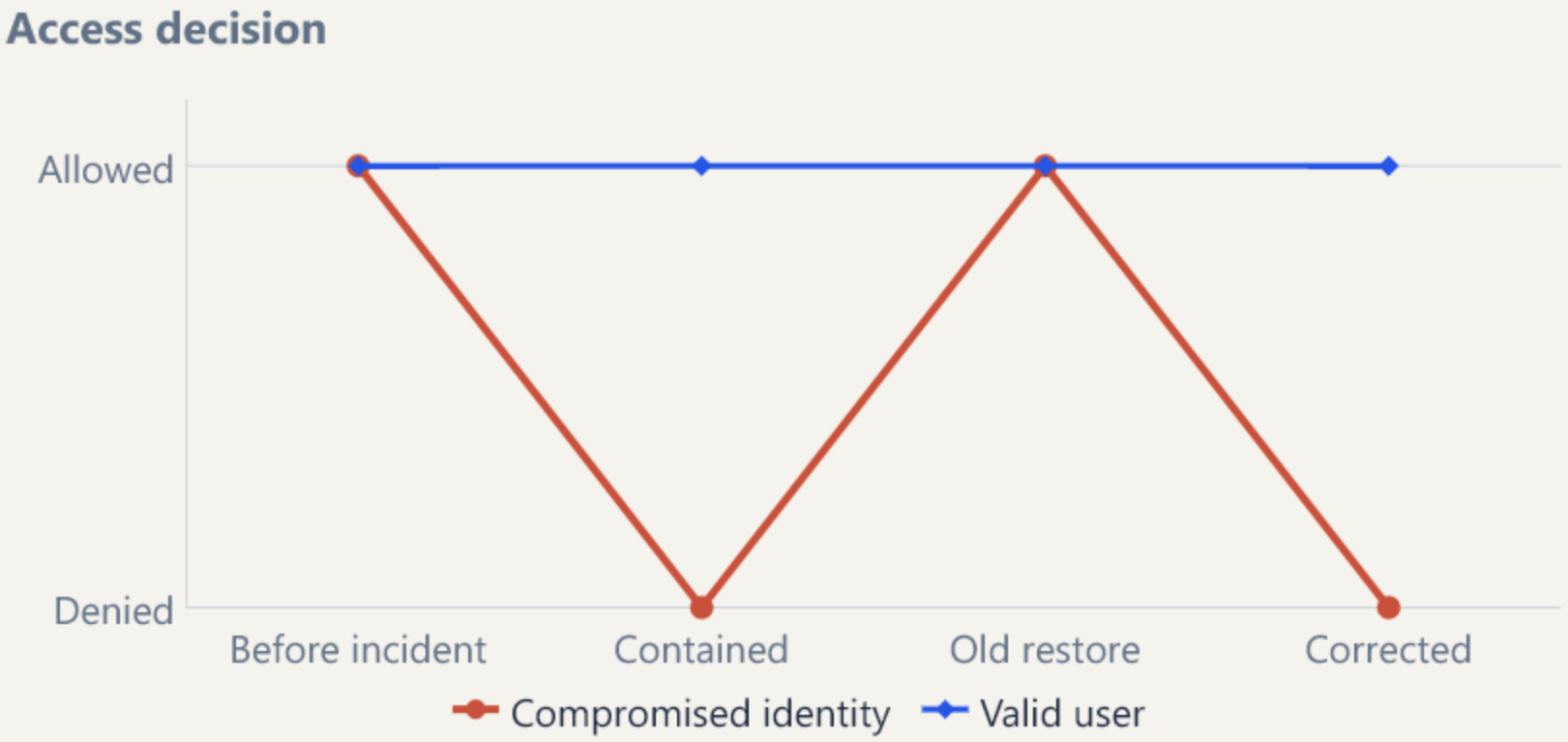
Containment verification after application recovery

Verifying that recovery preserves the security decisions made during an incident.



Team NullSynapse

Illustrative containment rollback scenario



Rollback

The old snapshot can restore access for the compromised identity.

The valid user should remain operational.

Assumption: account state lives in the restored application database.

Illustrative states, not measured results. Lines connect states and do not represent elapsed time.

Two conditions for an acceptable recovery

01

Business operation

A valid user authenticates and completes a test transaction.

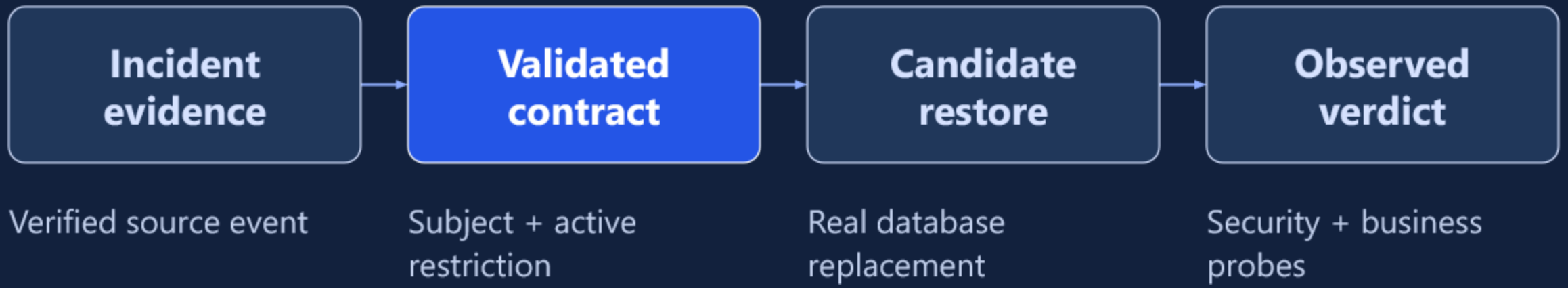
02

Containment preserved

The compromised identity remains denied under the active restriction.

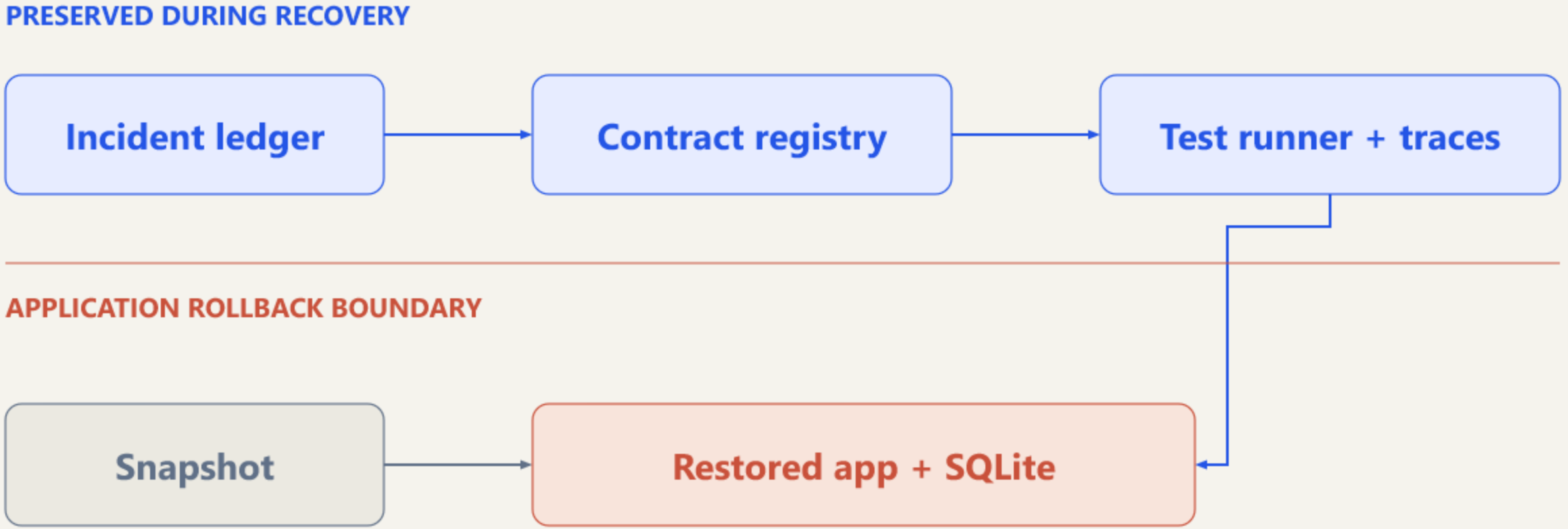
Both checks must hold. Denying every user fails the business condition.

Method: incident evidence to recovery tests



If a mandatory check fails Operator-approved correction, followed by the same tests

Architecture: independent verification state



Trust assumption: the restore cannot modify the ledger, accepted contracts or verifier.

Source-linked recovery requirements

A restriction becomes an executable assertion

The accepted requirement identifies the incident subject and approved event. A trusted SQLite adapter confirms account disablement alongside the HTTP access probe.

Source event	EVT-1042
Subject	svc-billing-7
Required behavior	Compromised access denied
Execution	Predefined HTTP probes
Validity	Until an approved later change

Credentials remain outside model prompts. The external ledger is a trusted prototype input.

AI interpretation with independent validation

01 Interpret

The optional Gemini adapter converts an incident narrative into a structured candidate requirement.

02 Validate

Deterministic rules check the source event, subject, supported probe and later approved changes.

03 Execute

Only validated requirements reach predefined tests. The model cannot generate or run arbitrary test code.

Current implementation

The offline rule baseline works without a key and is labelled as non-AI. The live adapter is implemented, but a live model call and AI accuracy evaluation remain unverified.

Verdicts follow observed evidence

FAIL	A validated mandatory check is false.
PASS	All required checks are true and evidence is complete.
UNVERIFIED	No known failure, but evidence or execution is incomplete.

PASS applies only to the nonempty declared check set. A timeout is not a successful denial test.

Observed restore failure and verified correction

FAIL / Old snapshot restores compromised access

Request	Identity	HTTP	Observed result	Time
POST /lab/login	svc-billing-7	200	Authenticated	1.5 ms
POST /lab/login	maya.rao	200	Authenticated	1.7 ms
POST /lab/invoices	maya.rao	201	INV-0001 created	3.4 ms

PASS / Approved correction preserves business access

Request	Identity	HTTP	Observed result	Time
POST /lab/login	svc-billing-7	403	Access denied	1.5 ms
POST /lab/login	maya.rao	200	Authenticated	1.3 ms
POST /lab/invoices	maya.rao	201	INV-0002 created	3.2 ms

Actual HTTP evidence from the prototype. PASS applies only to the declared checks.

Eight repeatable scenarios match expected verdicts

Scenario	Expected	Observed
Containment preserved	PASS	**PASS**
Old snapshot restored	FAIL	**FAIL**
Approved correction	PASS	**PASS**
Deny every user	FAIL	**FAIL**
Restrict the wrong account	FAIL	**FAIL**
Approved superseding event	PASS	**PASS**
Missing source evidence	UNVERIFIED	**UNVERIFIED**
Probe timeout	UNVERIFIED	**UNVERIFIED**

All 11 integration tests passed, including the eight-scenario matrix above. Synthetic results, not production accuracy.

The value of verifying containment after recovery

A focused check before returning an application to service

An application can be available while compromised access has returned. Blackout checks current incident restrictions alongside business operations, giving responders evidence for the recovery decision.

What makes the approach distinct

Incident-specific requirements survive rollback. Real access and business probes expose both restored attacker access and deny-all mistakes.

Next integrations

Connect trusted incident and identity systems, add application-specific probes, and evaluate live AI extraction against labelled records.

Scope: app-local account state. Verifier and ledger are trusted. Hashes show continuity, not tamper-proofing.

Blackout

Containment verification after recovery

A reproducible implementation of a focused recovery check: preserve incident intent, test the restored application and inspect the evidence before resuming service.

github.com/sidhesh0706/Blackout

Team NullSynapse

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